Markowitz Random Forest

Weighting decision trees with Modern Portfolio Theory

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1. Introduction & Motivation

The Traditional Random Forest Paradigm

Random Forest (RF) is an ensemble algorithm. By default, it combines individual decision trees through **uniform voting / averaging**

$$w_j = 1 / k \qquad (\forall\ j = 1, \ldots, k)$$

The core bottlenecks

- **Unequal Tree Quality:** As each tree is trained on different random data and features, some trees turn out much smarter and more reliable than others. Giving a weak tree and a strong tree the exact same voting power is inefficient.
- **The Diversity Dilemma:** Prior tree-weighting heuristics (e.g. SWRF, OWRF) assign weights strictly based on *individual tree accuracy*.
- **Why Individual Accuracy Fails:** High-performing trees often make *identical, correlated errors* on challenging edge cases. Weighting only the top trees destroys ensemble diversity and degrades generalization.

The Research Paper poses the question: Can we formulate tree weighting as an optimization problem that rewards accurate trees while explicitly penalizing trees that make the same mistakes?

2. The Proposed Idea: Portfolio Theory Analogy

Markowitz Modern Portfolio Theory (MPT / MVA)

Harry Markowitz (Nobel Laureate, 1952) showed that an optimal investment portfolio does not merely pick the highest-return stocks; it **diversifies risk** across uncorrelated assets to minimize overall volatility.

Finance to Machine Learning Mapping

Modern Portfolio Theory (Finance)	Markowitz Random Forest (Machine Learning)
Financial Assets	Individual Decision Trees in the Forest ($j = 1, \ldots, k$)
Asset Weights (w)	Tree Voting / Aggregation Weights ($w_j \geq 0, \sum w_j = 1$)
Expected Return (μ)	Out-of-Bag (OOB) Performance Score (r_j)
Asset Covariance Matrix (Σ)	Pairwise Tree Mistake Risk Matrix (K)
Portfolio Risk ($w^T \Sigma w$)	Joint participation of trees making correlated mistakes

A tree that is slightly less accurate individually can receive substantial weight if its mistakes are uncorrelated with the rest of the ensemble.

3. Proposed Architecture: Risk & Return Formulation

1. Quadratic Programming Optimization

$$\max_w [\, w^T r - \lambda\, w^T K\, w - \gamma\, \|w\|_2^2 \,] \quad \text{s.t.} \quad \sum w_j = 1, \quad 0 \le w_j \le 1$$

2. Expected Performance / Return Vector ($r \in R^k$)

Evaluated strictly on **Out-of-Bag (OOB)** samples to prevent in-sample bias:

• *Binary:* PR-AUC • *Multiclass:* Macro F1 • *Regression:* $r_j = \exp(-\text{MAPE}_j) \in [0, 1]$

3. The Risk Matrix (K)

In finance, risk is the covariance of asset returns—diversification fails if all stocks crash together. In MRF, Risk is correlated mistakes.

Regression (Residual Covariance): We calculate the covariance of the signed residuals ($y_i - \hat{y}_{ij}$). If Tree A and Tree B both overshoot or both undershoot the target on the exact same data points, their covariance is high, signaling redundant, correlated failure modes.

Binary (Mistake Subsets): Instead of measuring raw prediction correlation across the whole dataset, we look strictly at the mistake subsets: instances where the probability assigned to the correct label dropped below 0.5. We compute the correlation across False Positives and False Negatives separately, then average them.

Multiclass (Class-wise Mistake Correlation): Generalizes the binary idea: for each class c, filter only the samples where mistakes occurred, measure how strongly the trees' predicted probability vectors correlate on those mistakes, and average across all classes C.

4. Results: Classification Tasks

Multiclass

Macro F1 (Higher is Better ↑)

Dataset	Model	Paper	Replicated
pen (10 cls)	Standard RF	0.9819	0.9828
	IVRF (Variance)	0.9819	0.9828
	MRF (Proposed)	**0.9823**	**0.9829**
	XGBoost	0.9809	0.9898
student (3 cls)	Standard RF	0.6988	0.7059
	IVRF (Variance)	0.6988	0.7064
	MRF (Proposed)	**0.6993**	**0.7063**
	XGBoost	0.6685	0.7005

Binary

PR-AUC (Higher is Better ↑)

Dataset	Model	Paper	Replicated
shoppers (imbalance)	Standard RF	0.7244	0.7493
	IVRF (Variance)	0.7312	0.7498
	MRF (Proposed)	**0.7448**	**0.7504**
	XGBoost	0.7194	0.7550
bank_mkt (recall)	Standard RF	0.6280	0.5968
	IVRF (Variance)	0.6291	0.5968
	MRF (Proposed)	**0.6474**	**0.5961**
	XGBoost	0.6590	0.6257

5. Results: Regression Tasks

Regression

Metric: MAE (Lower is Better ↓)

Dataset	Model	Paper	Replicated
wine (quality)	Standard RF	0.5253	0.5425
	IVRF (Variance)	0.5252	0.5425
	MRF (Proposed)	**0.5242**	**0.5410**
	XGBoost	0.5537	0.5381
supercond. (temp Tc)	Standard RF	7.1781	6.5900
	IVRF (Variance)	7.1896	6.6473
	MRF (Proposed)	**7.1624**	**6.4273**
	XGBoost	7.5077	5.8413

Observations :

1. Covariance Beats Variance ($K > \sigma^2$)

IVRF (penalizing individual tree variance alone without pairwise covariance **K**) consistently fails to beat MRF. Joint error correlation is better suited for ensemble diversification.

2. Consistent Top Rank Across All Forests

In both the original paper and our 20-trial replication, **MRF achieves the lowest MAE among all Random Forest variants** across both regression and classification datasets.

6. Conclusions & Takeaways

- **Modern Portfolio Theory Translates to Ensembles:** Applying Markowitz mean-variance optimization to decision trees effectively balances individual tree accuracy with error diversification.
- **Mistake Covariance is Essential:** Comparing MRF against IVRF confirms that joint error correlation (K) produces superior ensembles compared to evaluating tree variance in isolation.
- **Consistent Gains over Standard RF:** Across classification and regression tasks, MRF consistently improves upon the uniform Random Forest (w = 1/k).
- **Plug-and-Play Post-Processing:** MRF optimizes pre-trained trees via convex quadratic programming (CVXPY) without requiring alterations to the underlying tree-growing algorithm.
- **Replication Verified:** Our 20-trial empirical replication closely reproduces the paper's reported numbers across tasks.